Linear Algebra
MATH 310 Section 003

SYLLABUS

Virginia Commonwealth University

Fall 2026
MWF 12:00-12:50
Harris Hall Room 3142

Instructor: Richard Hammack
E-mail: rhammack@vcu.edu
Course webpage: RichardHammack.github.io/Math310
YouTube: @StraightEdgeAndCompass

Office: Harris Hall 4166
Office Hours: MWF 10:50-11:50
MF 1:00-1:50
and by appointment

Linear Algebra is the mathematics of systems of equations involving many variables, none of which are raised to a power greater than 1. This course covers systems of linear equations, vector spaces, linear dependence, bases, dimension, linear mappings, matrices, determinants, quadratic forms, orthogonal reduction to diagonal form, eigenvalues and geometric applications.

Prerequisites: One of MATH 139, MATH 141, MATH 151, MATH 200 or MATH 201 (with a minimum grade of C) or a score on the VCU Mathematics Placement Test that places you into MATH 151 or higher.

Required Text: You may use any one of the following texts. Note that VCU's Bookstore stocks only the latest (8th) edition.
Elementary Linear Algebra, **Third** Edition, by Larson and Edwards (ISBN 0-669-39641-9)
Elementary Linear Algebra, **Fifth** Edition, by Larson, Edwards and Falvo (ISBN 0-618-33567-6)
Elementary Linear Algebra, **Sixth** Edition, by Larson and Falvo (ISBN: 0-618-78376-8)
Elementary Linear Algebra, **Seventh** Edition, by Larson (ISBN: 1-133-11087-8)
Elementary Linear Algebra, **Eighth** Edition, by Larson (ISBN: 1-305-65800-0)

It does not matter which of these editions you get. They are virtually the same, except that the exercises get renumbered and earlier editions are less expensive. Look for good prices online. Material from Chapters 1, 2, 3, 4, 6 and 7 is covered. This includes solving and interpreting systems of equations, matrix algebra, determinants, vector spaces, subspaces, linear transformations, eigenvalues and eigenvectors, and diagonalization.

Optional Solutions Manual: Depending on the edition you use, you may wish to get a student solution manual. (Some may be online.)
Student Solutions Manual for **Fifth** Edition, by Larson (ISBN: 0-618-33568-4)
Student Solutions Manual for **Sixth** Edition, by Larson (ISBN: 0-618-78377-6)
Student Solutions Manual for **Seventh** Edition, by Larson (ISBN-13: 978-1-13311132-0)
Student Solutions Manual for **Eighth** Edition, by Larson (ISBN: 1-305-65802-7)

Your grade is determined by scores on quizzes, tests, participation and a final exam.

Quizzes: There will be a short quiz most weeks:

- Each quiz closed-book and closed-notes.
- Use of calculators or other computing technology is not allowed on quizzes.

- No electronic devices of any type (other than a standard watch) may be used during any quiz.
- Several of your lowest quiz grades will be dropped.
- All quizzes are written under the assumption that you are studying the material ***at least 6 hours per week outside of class.***
- Your three lowest quiz grades are dropped.

Tests: There are two 50-minute tests, as scheduled on the calendar. Tests are in-class, closed-book, closed-notes and no technology is used. Your lowest test grade is dropped.

Participation: This means that you in some way demonstrate intellectual involvement in the course. It does not necessarily mean that you ask questions and volunteer answers. Active participation may include working lots of exercises and displaying preparedness, dedication and intellectual curiosity.

Final Exam: The final exam is comprehensive, covering material from chapters 1, 2, 3, 4, 6 and 7. It is closed-book and closed-notes. It is scheduled for 12:30-3:20 p.m. on Tuesday December 15. In writing the final exam, I will assume that you have been studying the material ***at least 6 hours per week outside of class, throughout the entire semester.***

Grading:

The 10-point grading scale is used:

A:	90-100
B:	80-89
C:	70-79
D:	60-69
F:	0-59

Your final average will be computed as follows:

Quizzes:	32%
Highest Test Grade:	32%
Participation:	4%
Final Exam:	32%
Total:	100%

Attendance: You are responsible for all material covered in class. I do not take attendance, but I do notice if you are not attending class. If your grades are high, I do not mind if you miss class occasionally; otherwise, excessive absences may result in a reduced participation score. As a matter of courtesy, you should arrive punctually and stay for the entire duration of each class you attend. Please inform me ahead of time if you must leave early.

I may administratively drop or withdraw from the course any student who has not attended class for four consecutive weeks without explanation.

Last day to withdraw: Friday, October 30.

Important: All exams and quizzes are **on campus, in-class and in-person.**

Notice: If I have questions about the work on any quiz or test, I may request a meeting to discuss the issues before returning the graded work.

Make-up Work: Under normal circumstances I do not give make-up quizzes and tests. I will drop three low quiz grades and your lowest test grade. If you miss a quiz or test, then that counts as a dropped grade (up to the number of grades dropped). If you miss the

final exam for a legitimate reason (i.e. a documented illness or emergency) then, following VCU policies, you can request a grade of Incomplete (I) for the course, and you will need to make up the missed exam by the date set by the University.

Generative AI: Neither the students nor the instructor will use generative AI in this course.

Recordings and Photography: Because of privacy considerations for all class members, recording video or audio in class is not allowed, except with documented approval by VCU's SAEO office. Please do not take photos in class.

Devices: While in class, you are expected to be connected with the course. All devices not related to coursework (phones, etc.) should be muted and put away for the entire duration of any class. **Smartwatches or Wearables that are voice or data connected (i.e. iOS and/or Android enabled devices such as an Apple Watch) are not permitted during quizzes and tests. Violation will be considered an honor offense.**

Web Page: Information about this course is posted on my web page (not on Canvas). Go to RichardHammack.github.io/Math310/. There you will find the syllabus, the course calendar, a test archive and other materials. Solutions for all graded work will be posted after the due dates.

Email: Any email correspondence concerning this course should be through your official VCU email address. I may send email messages either to the whole class or individuals in the class. It is your responsibility to check your VCU email regularly.

Office: Please feel free to stop by my office whenever you have a question, or if you just want to chat. If my posted hours are inconvenient, I will be happy to schedule an appointment. Tell me if you are having trouble. Catching up can be very difficult once you get behind, so let me know as soon as you think there is a problem.

Exercises: You should do as many of the following odd-numbered exercises as you can for practice. Answers to most are in the back of your text. Some of these will be used for the in-class tests and exams. Notice that the numbers of the exercises differ depending on which edition of the text you are using, but the actual problems are mostly the same. On the course web page you will also find full solutions to selected even-numbered problems.

Section	**Exercises**				
	Third Edition	*Fifth Edition*	*Sixth Edition*	*Seventh Edition*	*Eighth Edition*
1.1	1-29, 37-55 (odd)	1-45 (odd)	1-29, 37-55 (odd)	1-29, 37-55 (odd)	1-29, 37-55 (odd)
1.2	1-35	1-31, 39-61	1-35, 43-61	1-37, 41-51	1-37, 43-51
2.1	1-50	1-17, 21-37	1-17, 21-53	1-55	1-57
2.2	1-35	1-25, 31, 35-39	1-25, 35-41	1-33	1-45, 1-53
2.3	1-17, 21, 27-41	1-19, 25, 27, 31-37	1-19, 25, 27, 33-41	1-23, 41-47	1-25, 31-45
2.4	1-41	1-27	1-31	1-33	1-35
3.1	1-43	1-33	1-33	1-33	1-33, 39, 41
3.2	1-33	1-39	1-47	1-41	1-41

3.3	1-27, 31-39	1-13, 19-31, 35-43	1-17, 23-41, 45-53	1-33, 37-43, 53, 55	1-33, 37-49, 57
4.1	1-39	1-21, 25-37, 41-53	1-31, 35-49, 55-67	1-29, 33-47, 51-59	1-31, 35-49
4.2	1-25	1-37	1-41	1-39, 43-47	1-43
4.3	1-29	1-31	1-47	1-49, 53-59	1-45, 49-55
4.4	1-55	1-59	1-69	1-79	1-77
4.5	1-57	1-65	1-85	1-77	1-83
4.6	1-21, 25, 27, 31-39	1-23, 27, 29, 33-45, 49-53	1-65	1-65	1-45, 49-53, 57-61
4.7	1-19, 23-31	1-19, 23, 25, 29-35	1-17, 29-31, 35-47	1-23, 43-53	1-31, 37, 39, 45-51
6.1	1-39, 47, 49	1-41	1-57	1-65	1-59
6.2	1-41	1-19, 23-47	1-27, 31-55	1-27, 31-59	1-29, 33-59
6.3	1-45	1-19, 25-59	1-29, 39-73	1-19, 27-65	1-21, 27-43
6.4	1-25	1-17	1-17	1-21	1-23
7.1	1-51	1-27, 39-55	1-27, 51-65	1-27, 39-41, 51-55	1-27, 41-49
7.2	1-33	1, 3, 5, 7, 9, 13-27	1-13, 17-51	1-5, 7-19, 23-39	1-5, 7-19, 23-39

Boilerplate: **Use VCU Libraries** to find and access library resources, spaces, technology and services that support and enhance all learning opportunities at the university. (https://www.library.vcu.edu/) **Read VCU's syllabus statements at** https://faculty.provost.vcu.edu/faculty-resources/syllabus-resources/for-students//